

VisionDriveAI

AIoT Distraction Detection System for Urban Motorbikes

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Tập trung hơn, lái xe an toàn hơn.

Agenda

- 1 Problem and Market Context
- 2 Product Vision
- 3 Architecture
- 4 Hardware Prototype
- 5 Objectives and Use Cases
- 6 User Flow
- 7 Business Value
- 8 Roadmap

Motorbike riders need real-time attention support

In urban traffic, distraction can happen in seconds: drowsiness, looking away, phone usage, unstable handlebar motion.

- ▶ Motorbikes are common in Vietnam, but driver monitoring systems mostly target cars.
- ▶ Smartphone safety apps can track routes, but phones themselves can cause distraction.
- ▶ Dashcams record incidents, but often do not prevent them in real time.
- ▶ Riders need a low-cost, edge-first system with immediate feedback.

VisionDriveAI is an AIoT safety assistant for urban motorbike riders.

- ▶ Detect drowsiness, gaze deviation, and phone holding.
- ▶ Combine camera and IMU for stronger context.
- ▶ Alert locally through vibration, buzzer, and LED.
- ▶ Log trips and personalize model behavior over time.

VisionDriveAI

Camera + IMU

Edge AI

Realtime Alert

Urban Motorbike Safety

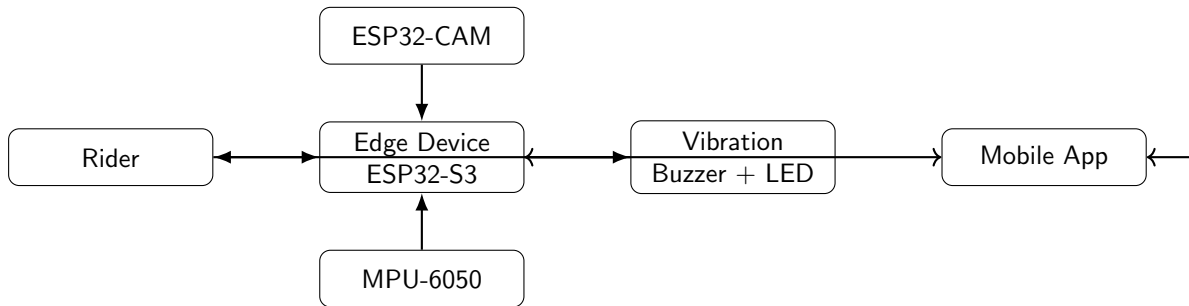


Figure: Edge-Mobile-Server Architecture

Hardware Prototype

Component	Role	Estimated Price
ESP32-S3	Edge inference, Wi-Fi/BLE, sensor fusion	150k
ESP32-CAM	Face, gaze, and hand/phone detection	80k
MPU-6050	Handlebar motion, braking, instability sensing	30k
Vibration Motor	First-level haptic warning on handlebar	20k
Buzzer + LED RGB	Escalated audio/visual alert	15k
LiPo + charging module	Independent prototype power	80k

Prototype cost

Total estimated hardware cost: **approximately 375,000 VND**

3 Objectives

Objective 1

Detect

- ▶ Drowsiness
- ▶ Gaze deviation
- ▶ Phone holding

Objective 2

Alert

- ▶ 3-level alert
- ▶ App notification
- ▶ AI pattern

Objective 3

Improve

- ▶ Safety Score
- ▶ Trends
- ▶ Personalization

Overall Use Cases

Objective	Use Case	AI	Core Value
Detect	Drowsiness and gaze deviation	Yes	Early attention risk detection
Detect	Phone holding while riding	Yes	Reduce phone-related distraction
Alert	Three-level escalation alert	No	Immediate local feedback
Alert	Realtime notification and pattern AI	Yes	Long-term risk insight
Improve	Trip Safety Score	No	Behavior feedback after ride
Improve	Personalized driver model	Yes	Fewer false positives

Objective 1: Real-Time Detection

Drowsiness + Gaze

- ▶ Eye closure over threshold.
- ▶ Head nodding down.
- ▶ Looking away continuously.

Phone Holding

- ▶ Detect hand-phone posture.
- ▶ Combine IMU to reduce false positives.
- ▶ Prioritize moving vehicle context.

Target latency: under 500ms on edge device

Objective 2: Alert and App Feedback

3-level escalation

- ① Light handlebar vibration.
- ② Strong vibration + short buzzer beep.
- ③ Continuous buzzer + red LED.

App and AI pattern

- ▶ Log event timestamp and type.
- ▶ Push notification for severe events.
- ▶ Detect risky time/route patterns.
- ▶ Adjust sensitivity over time.

Objective 3: Trip Analytics

Safety Score 0–100

After each trip, the app summarizes distraction events, eye-closure duration, phone holding, hard braking, and alert frequency.

- ▶ Weekly/monthly trend dashboard.
- ▶ Clear behavior feedback for riders.
- ▶ Personalized baseline after 1–2 weeks of data.
- ▶ Lower false positives compared with fixed thresholds.

Mount Device → Pair App → Start Trip → Detect + Alert → Safety Score

- ▶ Edge Device can alert without Internet.
- ▶ App enables logs, push notifications, and trip dashboard.
- ▶ Server enables pattern analysis and personalization.

- ▶ **Safety:** detects risky behavior before it becomes an incident.
- ▶ **Affordability:** prototype under 400k VND component cost.
- ▶ **Local relevance:** designed for Vietnam's motorbike-heavy urban traffic.
- ▶ **Scalability:** useful for families, delivery riders, and small fleets.
- ▶ **Data advantage:** long-term patterns can guide safer habits.

Limitations

- ▶ Low light and rain affect camera.
- ▶ Edge AI has memory/compute limits.
- ▶ Privacy needs careful handling.
- ▶ Real-road calibration is required.

Roadmap

- ▶ Robust enclosure and mounting.
- ▶ Better sensor fusion.
- ▶ Face/hand model optimization.
- ▶ Map-based risk hotspot analytics.
- ▶ Personalized driver profiles.

Thank You

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